Large parsimony problem

The large parsimony problem

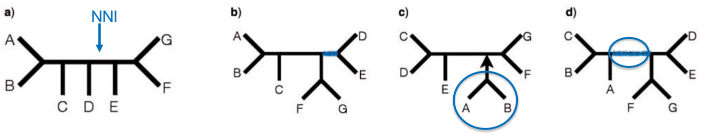
Input: A n x m matrix M describing n species, each represented by a word of m characters

Output: A tree T with n leaves labeled by the n lines of the matrix M, and a labeling of internal nodes such that the score of parsimony is minimized on all possible trees and all possible labeling of internal nodes.

- The search space is gigantic, and in particular when n gets larger
  - $(2n-3)!!$ rooted trees
  - $(2n-5)!!$ unrooted trees
- The problem is NP-complete

  An exhaustive search is possible but for very small values of n (< 10)
- Hence, branch and bound or heuristic algorithms are used

They are algorithms involving branch rearrangements



The starting tree

The interchange of neighboring subtrees yields a tree one Nearest Neighbor Interchange (NNI) move away

A Subtree Prune and Regraft (SPR) move: the subtree (A,B) is pruned from the initial trees and reattached

A Tree Bisection and Reconnection (TBR) move: the edge separating ABC from DEFG is bisected and reconnected by a new edge

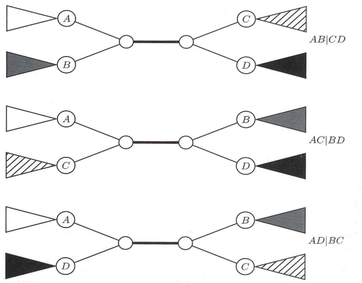
K.St John. The shape of phylogenetic treespaces, *Syst. Biol.* 2016

Nearest Neighbor Interchange : a greedy algorithm

It is an algorithm that rearranges the branches of the tree

- It evaluates only a subset of possible trees
- It defines a *neighborhood* of a tree as the one that can be reached by *exchanging the most close neighbor :*
  - It is the rearrangement (to be defined) of four subtrees determined by an internal edge.
  - Given a rule of manipulation of a subtree, there exist only 3 arrangements that are different for each edge.

Nearest Neighbor Interchange



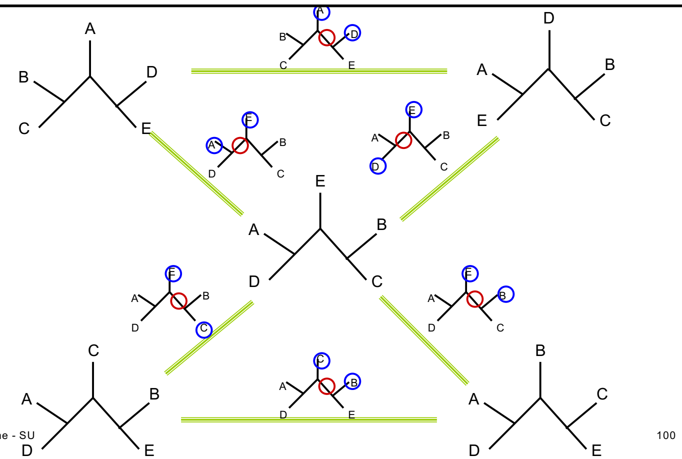
Algorithm : Nearest Neighbor Interchange

- Start with an arbitrarily chosen tree and test its neighbors
- Move on a neighbor if it gives a better score
- There is no way to know if the result is the most parsimonious tree
- It can stop in a local optimum

Possible neighboring relation :

Two trees are neighbors if they can be represented as quartets joining the same 4 subtrees.

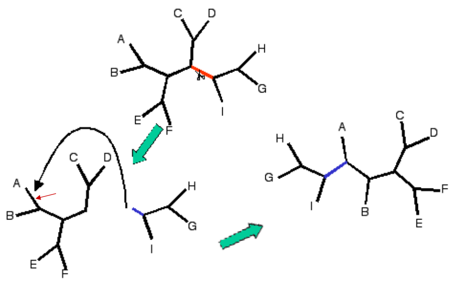
For instance:



The Subtree Prune and Regraft (SPR)

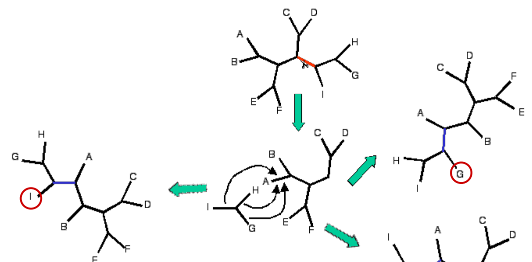
algorithm 2

Introduction of **one** new internal node for the second tree



Tree Bisection and Reconnection (BTR)

algorithm 3



We introduce **two** new internal nodes, one per tree

Problems with the phylogenetic reconstruction

- Important: a single phylogenetic reconstruction often provides an incomplete image of the reality.
- When several methods (parsimony, distance based, probabilistic) give the same result, then it is more probable to have reached the correct answer.

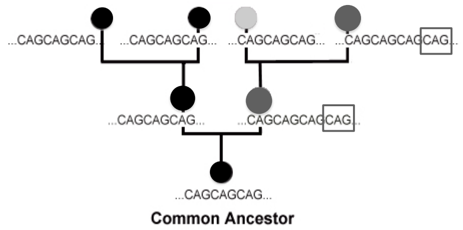
Homoplasy

- Given:
  - 1: ...CAGCAGCAG...
  - 2: ...CAGCAGCAG...
  - 3: ...CAGCAGCAGCAG...
  - 4: ...CAGCAGCAG...
  - 5: ...CAGCAGCAG...
  - 6: ...CAGCAGCAG...
  - 7: ...CAGCAGCAGCAG...

Sequences 1, 2, 4, 5 and 6 seem to be evolved from a common ancestor, with an insertion leading to 3 and 7

Homoplasy

- And if the true tree was this one?

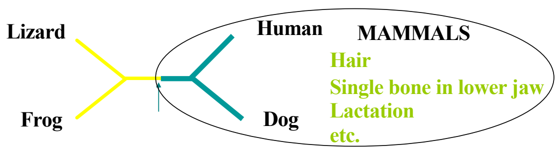


Homoplasy

- ● evolved separately compared to ●, but parsimony groups « ● » as having evolved from a common ancestor.
- Homoplasy: independent evolution (or parallel) of the same/similar character.
- Results of parcimony minimise homoplasy, in such a way that if homoplasy is frequent, parsimony can give very wrong results.

1. Contradictory characters: the case of tails

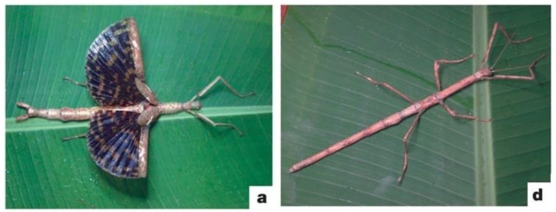
A phylogenetic tree has a strong probability of being correct when it is supported by several characters, as seen in this example:



Note: tails are homoplastic

2. How many times evolution re-invented the wings?

Whiting, et. al. (2003) looked at insects with and without wings.



Re-inventing the wings

- Previous studies demonstrated transitions “with → without” wings
- Transitions “without → with” wings are much more complicated (they ask for the development of several new biochemical pathways)
- Reconstruction of several phylogenetic trees induced always to establish re-evolution of wings.

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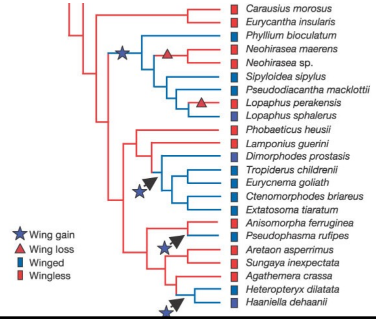
The most parsimonious phylogenetic tree of insects with and without wings



Phylogenetic tree based on **two criteria:** DNA sequences and presence/absence of wings

The most parsimonious reconstruction gives an ancestor with no wings

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Why insects without wings still fly?

Since the most parsimonious phylogenetic reconstruction ask for re-inventing wings, it is possible that the pathways of development for wings are conserved for insects with no wings.

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